

Fashion Product Image Classification with Convolutional Neural Networks

Final Project – Deep Learning · Prepared for: Head of Analytics

Executive Summary

- A CNN-based classifier was developed to automatically classify Fashion-MNIST products into 10 clothing categories.
- The recommended model achieved 92.98% test accuracy, compared with 88.34% for the baseline dense network.
- This represents approximately a 40% reduction in classification errors.
- Shirt was the most difficult category, with 77.1% recall and frequent confusion with T-shirt/top, Coat and Pullover.
- 93.5% of products were predicted with at least 70% confidence, and those predictions were 95.9% accurate.
- A practical implementation would automatically classify confident products and send the uncertain 6.5% for manual review.

Main Objective

The objective of this project was to build a supervised deep learning image-classification model using Convolutional Neural Networks (CNNs) to automatically classify fashion products into ten categories. For an online retailer, an accurate classification system could reduce manual product tagging, improve consistency, and speed up the process of organizing large product catalogues.

Deep learning technique: supervised deep learning for multi-class image classification, using a fully connected neural network as a baseline and **Convolutional Neural Networks (CNNs)** as the main approach, trained with Keras/TensorFlow.

Data Description

The analysis used the Fashion-MNIST dataset, consisting of 70,000 grayscale clothing images at a resolution of 28×28 pixels. The data is divided into 60,000 training images and 10,000 test images across ten equally represented product categories: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot.

Each image contains 784 pixel-intensity features, while the target variable identifies the product category.

Item	Detail
Source	Fashion-MNIST (Zalando product images), loaded via <code>keras.datasets.fashion_mnist</code>
Size	70,000 images: 60,000 training, 10,000 test
Features	$28 \times 28 = 784$ greyscale pixel intensities per image (0–255)
Target	Product category (10 classes, 7,000 images each)

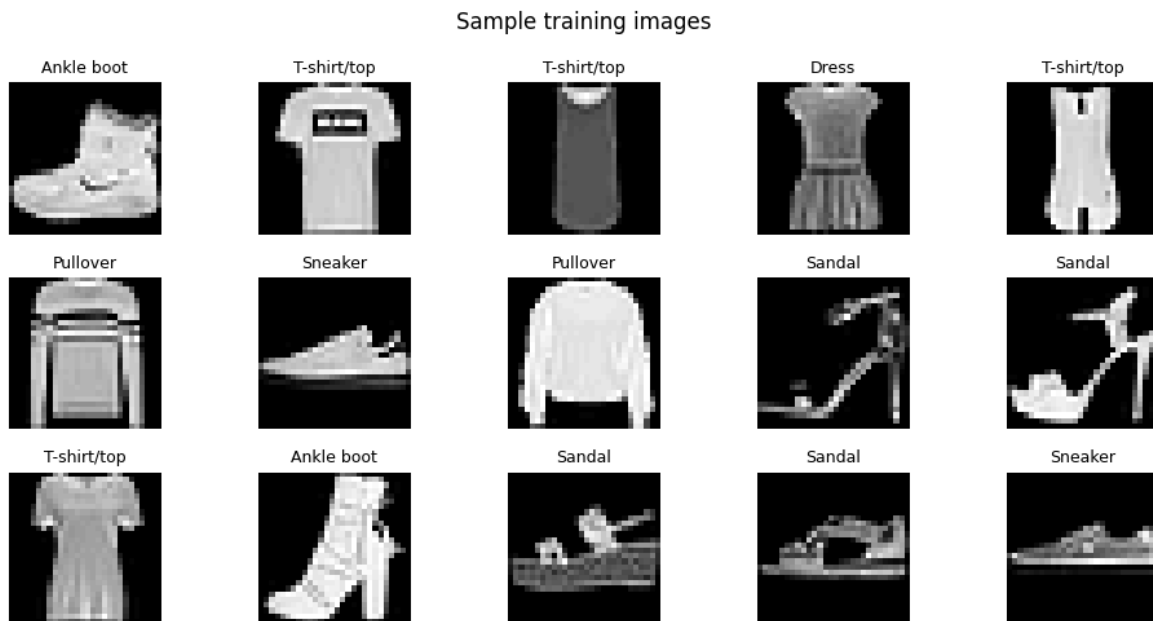


Figure 1. Sample training images with their category labels.

Data Exploration and Preparation

The dataset was perfectly balanced, with 6,000 training images in each category, meaning overall accuracy is a reasonable evaluation measure.

Pixel values originally ranged from 0 to 255 and were normalized to a 0–1 scale to improve neural-network training. A channel dimension was added to prepare the images for convolutional layers. During model development, 10% of the training data was reserved for validation.

Model Variations

Four neural-network configurations were evaluated.

The initial Dense Neural Network achieved 88.34% test accuracy, providing a baseline for comparison. Replacing the dense architecture with a CNN increased test accuracy to 90.93%, demonstrating the advantage of learning spatial patterns in images. However, the basic CNN showed clear overfitting, reaching 97.19% training accuracy while validation performance was substantially lower.

A regularized CNN using Batch Normalization, Dropout and Early Stopping reduced overfitting and improved test loss, although test accuracy remained around 90.95%.

The final model added a learning-rate reduction schedule and longer early-stopping patience. It produced the strongest overall performance:

Model	Architecture / training changes	Parameters	Epochs	Final train acc.	Best val. acc.	Test acc.	Test loss
1. Dense NN	Flatten → Dense(128) → Softmax	101,770	15	91.9%	88.8%	88.3%	0.346
2. Basic CNN	2 × (Conv 3×3 + MaxPool) → Dense(128)	421,642	15	97.2%	92.2%	90.9%	0.314
3. Regularised CNN	+ BatchNorm, Dropout, EarlyStopping (patience 5)	422,026	15	91.4%	91.6%	91.0%	0.269
4. Regularised CNN + LR schedule	+ ReduceLROnPlateau, patience 10	422,026	38	94.1%	93.3%	93.0%	0.225

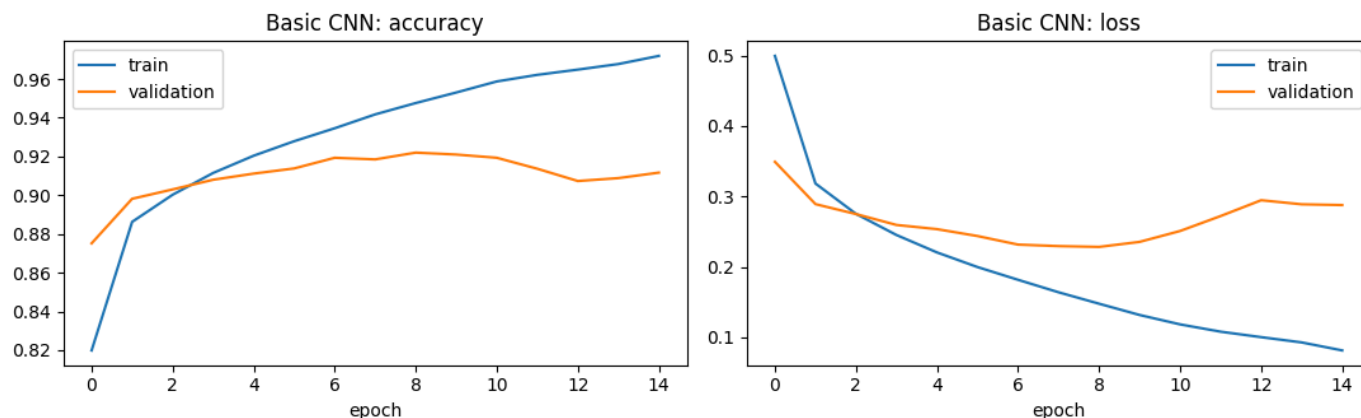


Figure 2. Basic CNN learning curves: training accuracy keeps rising while validation loss increases after epoch 9, a clear sign of overfitting.

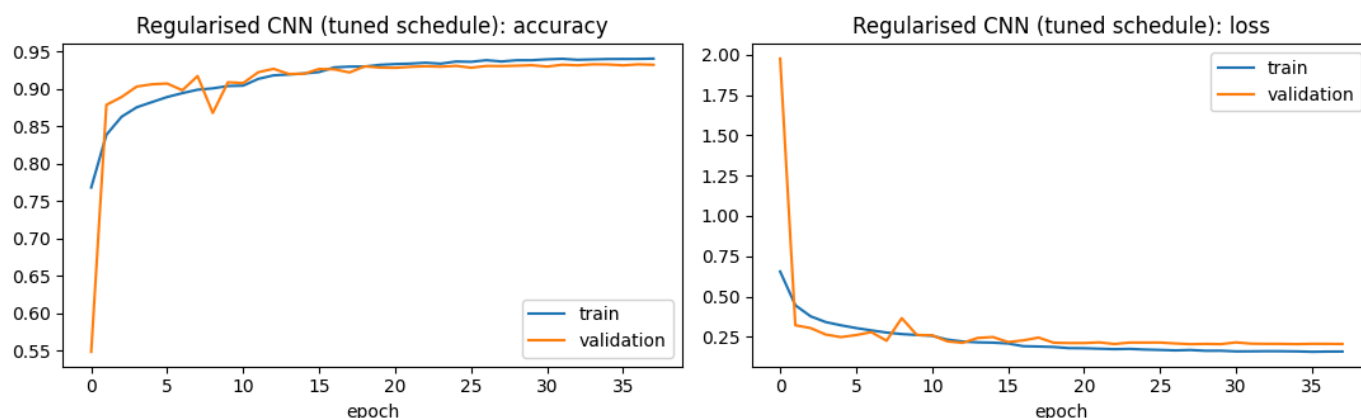


Figure 3. Final model learning curves: training and validation stay close; each learning-rate reduction is followed by further validation improvement.

Final Model Recommendation

The Regularised CNN with the learning-rate schedule is recommended as the final model. It achieved 92.98% test accuracy, a test loss of 0.2251, and a best validation accuracy of 93.30%.

Its final training accuracy was 94.06%, giving only a small gap between training and validation performance and suggesting substantially better generalization than the basic CNN.

One trade-off is model size. The final CNN contains approximately 422,000 parameters, compared with around 102,000 in the original dense model, meaning it requires more computation and training time. However, the improvement in predictive performance justifies the additional complexity.

Key Findings and Insights

The final CNN reduced the classification error rate from 11.66% with the dense network to 7.02%, representing an error reduction of approximately 40%.

Performance varied by product category. Trouser, Sandal, Bag, Sneaker and Ankle boot were classified particularly well, with recall between approximately 97% and 99%. Dress achieved 93.1% recall, while T-shirt/top, Pullover and Coat were around 90%.

The most difficult class was Shirt, with 77.1% recall and 78.1% precision. Of the 1,000 true Shirt images, only 771 were classified correctly. The largest errors were 87 Shirts classified as T-shirt/top, 56 as Coat, and 52 as Pullover. This suggests that visually similar upper-body garments remain difficult to distinguish at the low 28×28 resolution.

The model's confidence scores also provide a practical business opportunity. 93.5% of test images received predictions with at least 70% confidence, and those predictions were 95.9% accurate. Therefore, the retailer could automatically tag the confident 93.5% of products while sending the remaining 6.5% to human reviewers. This could remove more than 90% of manual classification work while maintaining high accuracy.

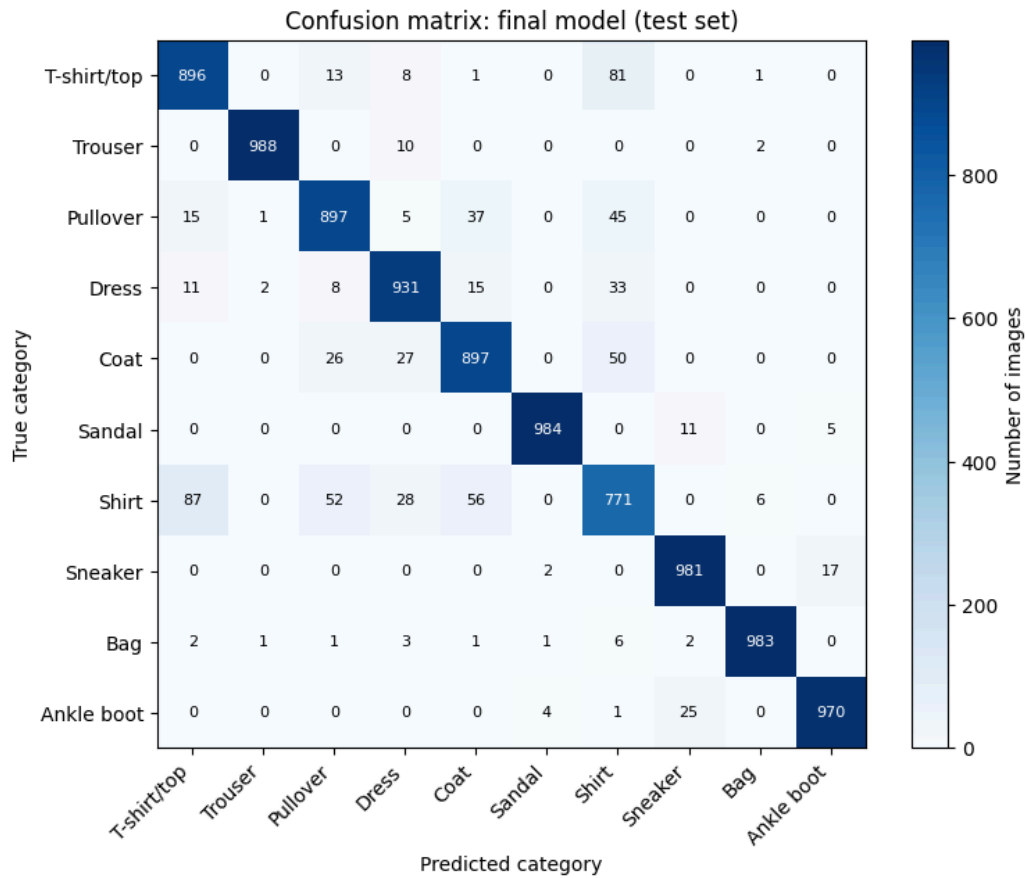


Figure 4. Confusion matrix of the final model on the 10,000 test images. Off-diagonal cells show which categories are confused; the largest errors are among the upper-body garments.

Limitations and Next Steps

The main limitation is that Fashion-MNIST contains small 28×28 grayscale images, which are much simpler than real online retail photographs containing colour, different backgrounds, lighting conditions and viewing angles.

Several visually similar clothing categories also remain difficult to distinguish, particularly Shirt, T-shirt/top, Pullover and Coat. In addition, model validation relied on a single validation split, and no image augmentation was used.

Future improvements could include data augmentation, higher-resolution colour images and transfer learning using pretrained CNNs such as MobileNet or ResNet. A specialised classifier could also be developed for the visually similar upper-body categories. Testing the system on real retailer images would be an important next step before practical deployment.

Appendix

The supporting notebook (`fashion_mnist_cnn.ipynb`) contains the full code for data loading, preparation, the four models, and the evaluation.